Next, we want to see the waterfall model. This means we want to visualize the temporal rendering of the service execution as all four of the services work together to serve a request. For that, click the Trace Results button in the upper-right corner of the dependency graph, and then select a trace (it doesn't matter which one; simply pick the top one in the list). You should then see something along the lines of figure 8.6. This screenshot shows an expanded single span that looks at the invocation of the service `B0`. From this, we can tell it took some 135 ms to execute, and of that, it spent more than 129 ms in the call to an external resource. Now that we've seen the happy path, let's simulate some failure and use Jaeger to figure out what's going on.

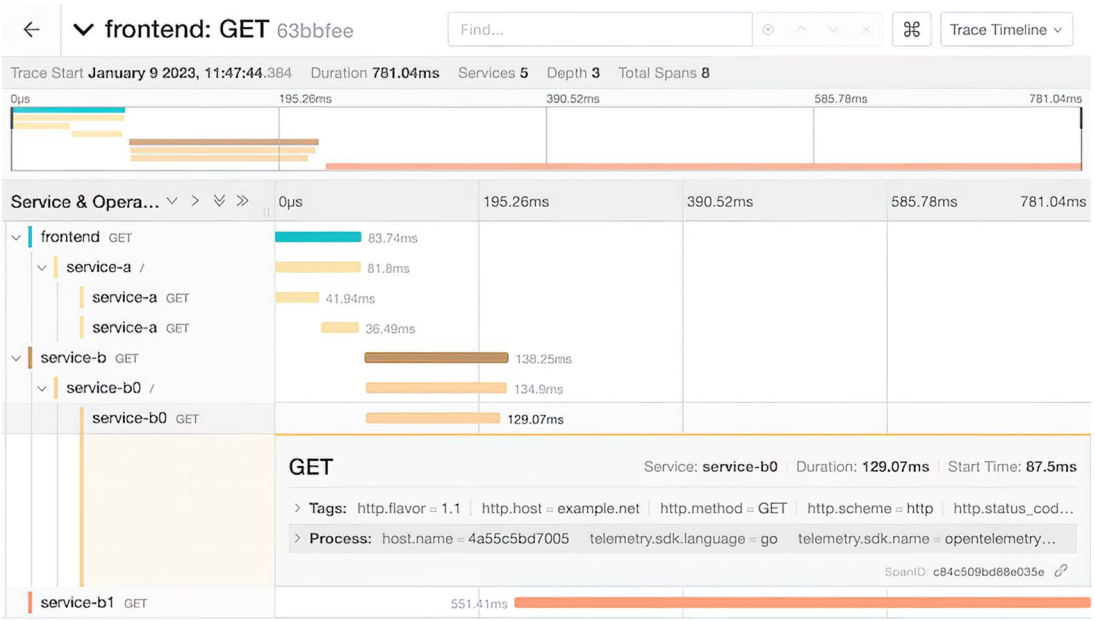


Figure 8.6 Troubleshooting a demo microservices app: a happy path example trace with its spans

8.2.4 Exploring a failure in the example app

To begin our failure example, stop the Docker Compose stack with CTRL-C, and to reset the app, also execute `docker-compose -f docker-compose.yaml down` so that all containers are removed. Now, go to the `ch08/jaeger/` directory and open the docker-compose.yaml file in your editor of choice, removing the comments (the `#`) in front of `HO11Y_CUTOFF_TPS` in line 62, as well as line 76 in front of `HO11Y_INJECT_FAILURE`. This causes services `B0` and `B1` to show errors.

Once you've saved your changes in the editor, do a `docker-compose -f docker-compose.yaml up` to bring up the stack, and go back to the Jaeger UI in your web browser of choice. There, again, select `frontend` in the left-hand-side box labeled Search, and then click the Find Traces button at the bottom.